Tutorial 5

1. Verify that y_1 and y_2 are solutions of the homogeneous equation $x^2y'' - 3xy' + 4y = 0$. To form a basis, the solutions must be linearly independent. We calculate the Wronskian $W(y_1, y_2)$:

$$\begin{aligned}
W(y_1, y_2) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \\
&= y_1y_2' - y_2y_1' \\
&= (x^2)(2x\log x + x) - (x^2\log x)(2x) \\
&= 2x^3\log x + x^3 - 2x^3\log x \\
&= x^3.
\end{aligned}$$

Since $W(y_1, y_2) = x^3 \neq 0$ for $x > 0$, the solutions are linearly independent. Because y_1 and y_2 are two linearly independent solutions to a second-order linear homogeneous differential equation, they form a basis for the solution space.
To find a solution of the non-homogeneous equation:

$$x^2y'' - 3xy' + 4y = x^2\log x, \quad x > 0 \tag{1}$$

We use the method of variation of parameters:

$$y_p(x) = -y_1\int \frac{y_2r(x)}{W(y_1, y_2)}dx + y_2\int \frac{y_1r(x)}{W(y_1, y_2)}dx \tag{2}$$

First, compute the Wronskian $W(y_1, y_2)$:

$$\begin{aligned}
W(y_1, y_2) &= y_1y_2' - y_2y_1' \\
&= x^2(x + 2x\log x) - (x^2\log x)(2x) \\
&= x^3
\end{aligned}$$

Using (2), we get:

$$y_p(x) = -x^2\int \frac{(\log x)^2}{x}dx + x^2\log x\int \frac{\log x}{x}dx$$

$$
\begin{aligned}
y_p &= -x^2 \int u^2\,du + x^2 \log x \int u\,du \quad \text{where } \begin{cases} u = \log x \\ du = \frac{1}{x} dx \end{cases} \\
&= -x^2 \frac{(\log x)^3}{3} + x^2 \log x \left(\frac{(\log x)^2}{2} \right) \\
&= \frac{x^2 (\log x)^3}{6}
\end{aligned}
$$

The general solution of (1) is

$$
y = C_1 x^2 + C_2 x^2 \log x + \frac{x^2 (\log x)^3}{6}.
$$

2. (a) $\sum_{m=0}^{\infty} m(m+1)x^m$
   Radius of convergence $R = \frac{1}{\lim_{m\to\infty} \left| \frac{a_{m+1}}{a_m} \right|}$

$$
\begin{aligned}
R &= \frac{1}{\lim_{m\to\infty} \left| \frac{(m+1)(m+2)}{m(m+1)} \right|} \\
&= \left[\lim_{m\to\infty} \left| \frac{m+2}{m} \right| \right]^{-1} = 1
\end{aligned}
$$

   (b) $\sum_{m=0}^{\infty} \frac{(-1)^m}{k^m} x^{2m} \quad \left(a_m = \frac{(-1)^{m/2}}{k^{m/2}} \right)$
   Radius of convergence $R = \frac{1}{\lim_{m\to\infty} |a_m|^{1/m}}$

$$
\begin{aligned}
R &= \frac{1}{\lim_{m\to\infty} \left| \frac{(-1)^{m/2}}{k^{m/2}} \right|^{1/m}} = \frac{1}{\lim_{m\to\infty} \frac{1}{\sqrt{k}}} \\
&= \sqrt{k}
\end{aligned}
$$

   (c)

$$
\begin{aligned}
R &= \frac{1}{\lim_{m\to\infty} \left| \frac{a_{m+1}}{a_m} \right|}, \quad a_m = \frac{1}{((2m+1)!)^{1/2}} \\
&= \frac{1}{\lim_{m\to\infty} \left| \frac{(2m+1)!}{(2m+3)!} \right|} \\
&= \frac{1}{\lim_{m\to\infty} \left| \frac{1}{(2m+3)(2m+2)} \right|} \\
&= \lim_{m\to\infty} (2m+3)(2m+2) \\
&= \infty
\end{aligned}
$$

3. (a)

$$(1+x^2)y'' - 4xy' + 6y = 0 \tag{3}$$

Let $y = \sum_{m=0}^{\infty} a_m x^m$ be a power series solution of (3). Then:

$$y' = \sum_{m=1}^{\infty} m a_m x^{m-1} \tag{4}$$

$$y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2} \tag{5}$$

$\Rightarrow (1+x^2)y'' = \sum_{m=2}^{\infty} m(m-1)a_m x^{m-2} + \sum_{m=2}^{\infty} m(m-1)a_m x^m$

$$= \sum_{m=0}^{\infty} (m+1)(m+2)a_{m+2}x^m + \sum_{m=2}^{\infty} m(m-1)a_m x^m$$

$$= 2a_2 + 6a_3 x + \sum_{m=2}^{\infty} \left\{(m+1)(m+2)a_{m+2} + m(m-1)a_m\right\} x^m \tag{6}$$

Using (4), we have:

$$\begin{aligned} -4xy' &= -\sum_{m=1}^{\infty} 4m a_m x^m \\ &= -4a_1 x - \sum_{m=2}^{\infty} 4m a_m x^m \end{aligned} \tag{7}$$

$$6y = 6a_0 + 6a_1 x + \sum_{m=2}^{\infty} 6a_m x^m \tag{8}$$

Putting the values from (6), (7), and (8) into (3) to get:

$2a_2 + 6a_0 + 2a_1 x + 6a_3 x +$

$$\sum_{m=2}^{\infty} \left\{(m+1)(m+2)a_{m+2} + m(m-1)a_m - 4m a_m + 6a_m\right\} x^m = 0$$

$\Rightarrow 2(3a_0+a_2)+2(a_1+3a_3)x+\sum_{m=2}^{\infty} \left\{(m+1)(m+2)a_{m+2} + (m-2)(m-3)a_m\right\} x^m = 0$

Comparing the coefficients of x^m for $m \geq 0$:

$$3a_0 + a_2 = 0 \Rightarrow a_2 = -3a_0$$
$$a_1 + 3a_3 = 0 \Rightarrow a_3 = -\frac{a_1}{3}$$

For the general term:

$$(m+1)(m+2)a_{m+2} + (m-2)(m-3)a_m = 0$$
$$\Rightarrow a_{m+2} = -\frac{(m-2)(m-3)}{(m+1)(m+2)} a_m \quad \ldots \text{ (Recurrence Relation)}$$

For $m = 2$:

$$a_4 = -\frac{0}{12}a_2 = 0 \Rightarrow a_6 = a_8 = \cdots = 0$$

For $m = 3$:

$$a_5 = -\frac{0}{20}a_3 = 0 \Rightarrow a_7 = a_9 = \cdots = 0$$

Using these coefficient values in $y = \sum_{m=0}^{\infty} a_m x^m$:

$$\begin{aligned} y &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + 0 + \dots \\ &= a_0 + a_1 x + (-3a_0)x^2 + \left(-\frac{a_1}{3}\right) x^3 \\ &= a_0(1 - 3x^2) + a_1 \left(x - \frac{1}{3}x^3\right) \\ &= a_0 y_2 + a_1 y_1 \end{aligned}$$

(b) For all real numbers x, the power series y_1 and y_2 converge since they are polynomials. Hence the radius of convergence $R = \infty$.

4.

$$(1 - x)y'' + xy' - y = 0, \quad y(0) = -3,\ y'(0) = 2 \tag{9}$$

Let $y = \sum_{m=0}^{\infty} a_m x^m$. Then:

$$xy' = \sum_{m=1}^{\infty} m a_m x^m \tag{10}$$

$$\begin{aligned} (1 - x)y'' &= \sum_{m=2}^{\infty} m(m-1)a_m x^{m-2} - \sum_{m=2}^{\infty} m(m-1)a_m x^{m-1} \\ &= \sum_{m=0}^{\infty} (m+1)(m+2)a_{m+2}x^m - \sum_{m=1}^{\infty} m(m+1)a_{m+1}x^m \\ &= \sum_{m=1}^{\infty} \{(m+1)(m+2)a_{m+2} - m(m+1)a_{m+1}\}x^m + 2a_2 \end{aligned} \tag{11}$$

Hence (9) becomes:

$$(2a_2 - a_0) + \sum_{m=1}^{\infty} \{(m+1)(m+2)a_{m+2} - m(m+1)a_{m+1} + m a_m - a_m\}x^m = 0$$

$$\Rightarrow a_2 = a_0/2$$

Recurrence Relation:

$$a_{m+2} = \frac{m}{m+2}a_{m+1} - \frac{(m-1)}{(m+1)(m+2)}a_m$$

For $m=1$: $a_3 = \frac{1}{3}a_2 = \frac{1}{3}(\frac{a_0}{2}) = \frac{a_0}{3!}$
For $m=2$: $a_4 = \frac{2}{4}a_3 - \frac{1}{12}a_2 = \frac{1}{2}(\frac{a_0}{6}) - \frac{a_0}{24} = \frac{a_0}{24} = \frac{a_0}{4!}$
For $m=k$: $a_{k+2} = \frac{a_0}{(k+2)!}$

Final Solution:

$$
\begin{aligned}
y &= a_1 x + a_0\{1 + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \dots\} \\
&= (a_1 - a_0)x + a_0 e^x \qquad (*)
\end{aligned}
$$

From (9), given initial conditions $y(0) = -3$ and $y'(0) = 2$:

- $y(0) = a_0 = -3$
- $y'(0) = a_1 - a_0 + a_0 = a_1 = 2$

Substituting $a_0 = -3$ and $a_1 = 2$ into the solution:

$$y = (2 - (-3))x + (-3)e^x = 5x - 3e^x$$

5. (a) Given the differential equation:

$$y'' - 4xy' + (4x^2 - 2)y = 0$$

Assume a power series solution of the form:

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Then its derivatives are:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting these into the differential equation:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 4x\sum_{n=1}^{\infty} n a_n x^{n-1} + 4x^2 \sum_{n=0}^{\infty} a_n x^n - 2\sum_{n=0}^{\infty} a_n x^n = 0$$

Adjusting the indices to have common powers of x^n:

- $y'' = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n$
- $-4xy' = \sum_{n=1}^{\infty} -4n a_n x^n$
- $4x^2 y = \sum_{n=2}^{\infty} 4a_{n-2}x^n$
- $-2y = \sum_{n=0}^{\infty} -2a_n x^n$

Combining coefficients for each power of x:

- **For x^0 ($n=0$):** $2a_2 - 2a_0 = 0 \implies a_2 = a_0$

- **For x^1 $(n = 1)$:** $6a_3 - 4a_1 - 2a_1 = 0 \implies 6a_3 = 6a_1 \implies a_3 = a_1$
- **General recurrence relation $(n \geq 2)$:**

$$(n+2)(n+1)a_{n+2} - (4n+2)a_n + 4a_{n-2} = 0$$

(b)

(c) Given the Legendre-type differential equation:

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

Substituting the power series $y = \sum_{n=0}^{\infty} a_n x^n$ and its derivatives:

$$(1 - x^2)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 2x\sum_{n=1}^{\infty} na_n x^{n-1} + 2\sum_{n=0}^{\infty} a_n x^n = 0$$

Expanding the terms:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=1}^{\infty} 2na_n x^n + \sum_{n=0}^{\infty} 2a_n x^n = 0$$

Shift indices for the first summation:

$$\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n - \sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=1}^{\infty} 2na_n x^n + \sum_{n=0}^{\infty} 2a_n x^n = 0$$

Grouping coefficients for x^n:

- **For x^0 $(n = 0)$:** $2a_2 + 2a_0 = 0 \implies a_2 = -a_0$
- **For x^1 $(n = 1)$:** $6a_3 - 2a_1 + 2a_1 = 0 \implies 6a_3 = 0 \implies a_3 = 0$
- **General recurrence relation $(n \geq 2)$:**

$$(n+2)(n+1)a_{n+2} - [n(n-1) + 2n - 2]a_n = 0$$

  Simplifying the bracket: $n^2 - n + 2n - 2 = n^2 + n - 2 = (n+2)(n-1)$.

$$(n+2)(n+1)a_{n+2} - (n+2)(n-1)a_n = 0$$

$$a_{n+2} = \frac{(n-1)}{(n+1)}a_n$$

Since $a_3 = 0$, all subsequent odd terms $a_5, a_7, \ldots$ will also be 0.

6. As the initial values are given at π, the expansion should be about $x_0 = \pi$. First introduce a new variable $t = x - \pi$. Then $x = t + \pi$ and the equation becomes:

$$y'' + (\sin t)y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Now write

$$y(t) = \sum_{n=0}^{\infty} a_n t^n$$

and substitute into the equation, recalling

$$\sin t = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1},$$

we reach

$$\begin{aligned}
0 &= y'' + (\sin t)y \\
&= \sum_{n=2}^{\infty} a_n n(n-1)t^{n-2} + \left(t - \frac{1}{3!}t^3 + \dots\right)(a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots) \\
&= 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 + \dots \\
&\quad + a_0 t + a_1 t^2 + \left(a_2 - \frac{a_0}{6}\right) t^3 + \dots \\
&= 2a_2 + (6a_3 + a_0)t + (12a_4 + a_1)t^2 + \left(20a_5 + a_2 - \frac{a_0}{6}\right) t^3 + \dots
\end{aligned}$$

This gives

$$\begin{aligned}
2a_2 &= 0 \\
6a_3 + a_0 &= 0 \\
12a_4 + a_1 &= 0 \\
20a_5 + a_2 - \frac{a_0}{6} &= 0
\end{aligned}$$

Applying the initial values we have

$$a_0 = 1, \quad a_1 = 0.$$

Thus the above leads to

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = 0, \quad a_3 = -\frac{1}{6}, \quad a_4 = 0, \quad a_5 = \frac{1}{120}.$$

The solution in the t variable is

$$y(t) = 1 - \frac{1}{6}t^3 + \frac{1}{120}t^5 + \dots$$

Returning to the x-variable, we have

$$y(x) = 1 - \frac{1}{6}(x-\pi)^3 + \frac{1}{120}(x-\pi)^5 + \dots$$